Lab Report 07

Course title: Computer Network Laboratory
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Submitted to-

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Title: Implementation of VLAN Using Three Switches

Objective:

- To understand the concept of VLAN.
- To create and configure three VLANs.
- To assign PCs to different VLANs.
- To configure access and trunk ports.
- To test communication between PCs in the same and different VLANs.

Description:

Three switches and 27 PCs were added to Cisco Packet Tracer. Nine PCs were connected to each switch and the three switches were interconnected using trunk links.

The PCs were configured with IP addresses and subnet masks according to their assigned VLANs. Three VLANs were created on all switches using the following names:

VLAN 10 - CSE
VLAN 20 - ICT
VLAN 30 - EEE

On each switch, ports Fa0/1-Fa0/3 were assigned to VLAN 10, Fa0/4-Fa0/6 to VLAN 20 and Fa0/7-Fa0/9 to VLAN 30. These ports were configured as access ports.

Ports Fa0/10 and Fa0/11 on each switch were configured as trunk ports to carry traffic from multiple VLANs between switches.

The show vlan brief and show interfaces trunk commands were used to verify the VLAN and trunk configurations.

Connectivity was tested using the ping command. PCs in the same VLAN could communicate across switches, while PCs in different VLANs could not communicate directly without inter-VLAN routing.

Connection:

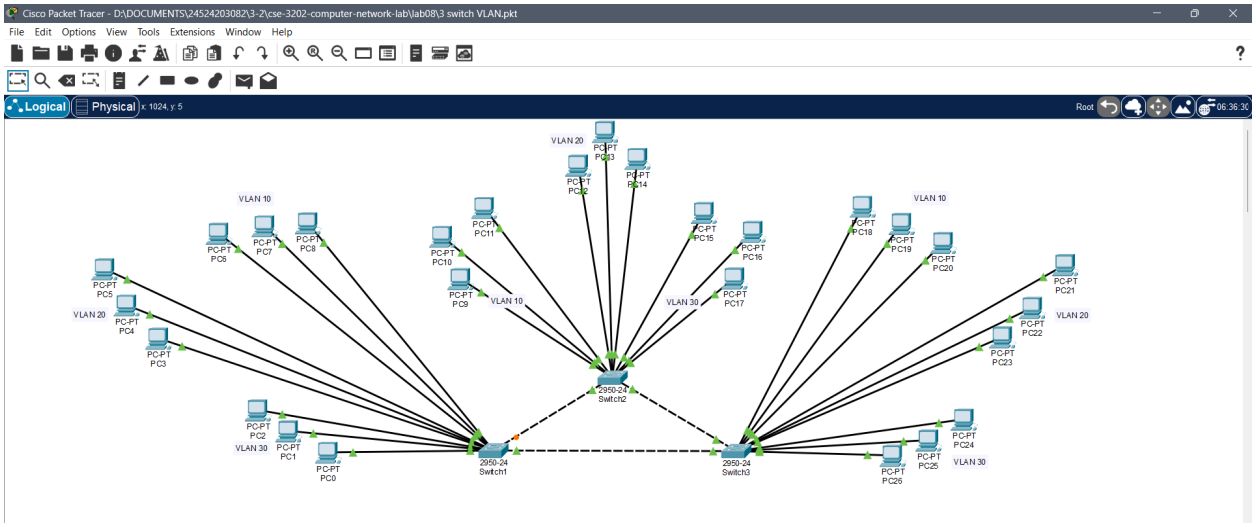


Figure 1: Complete Network Topology

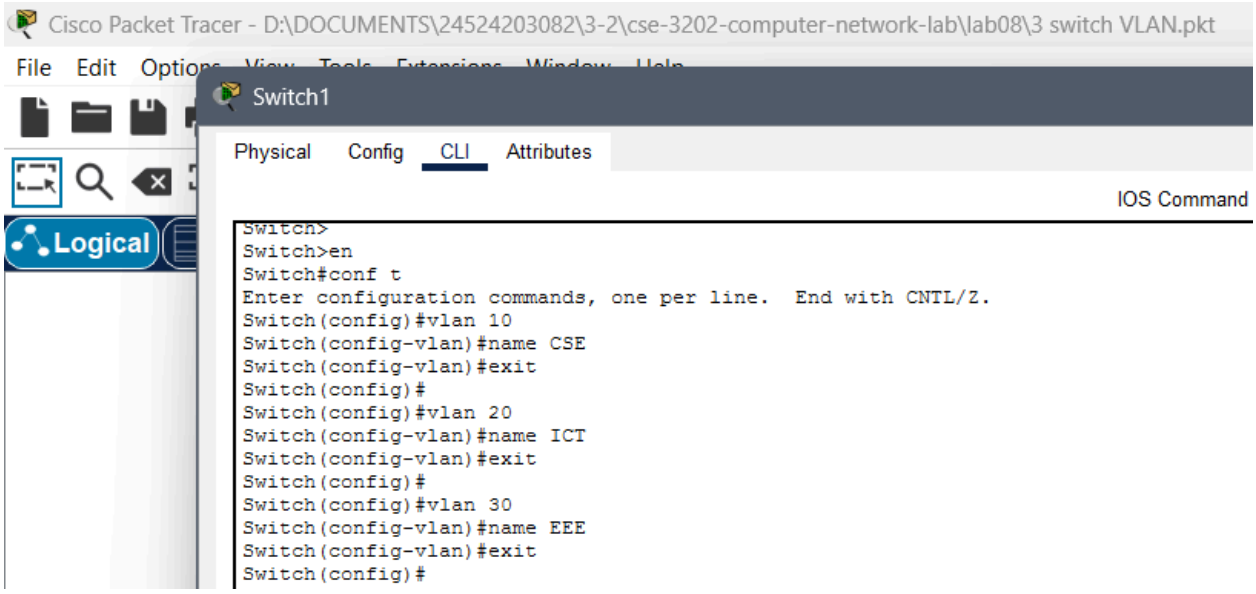


Figure 2: VLAN Creation and Naming

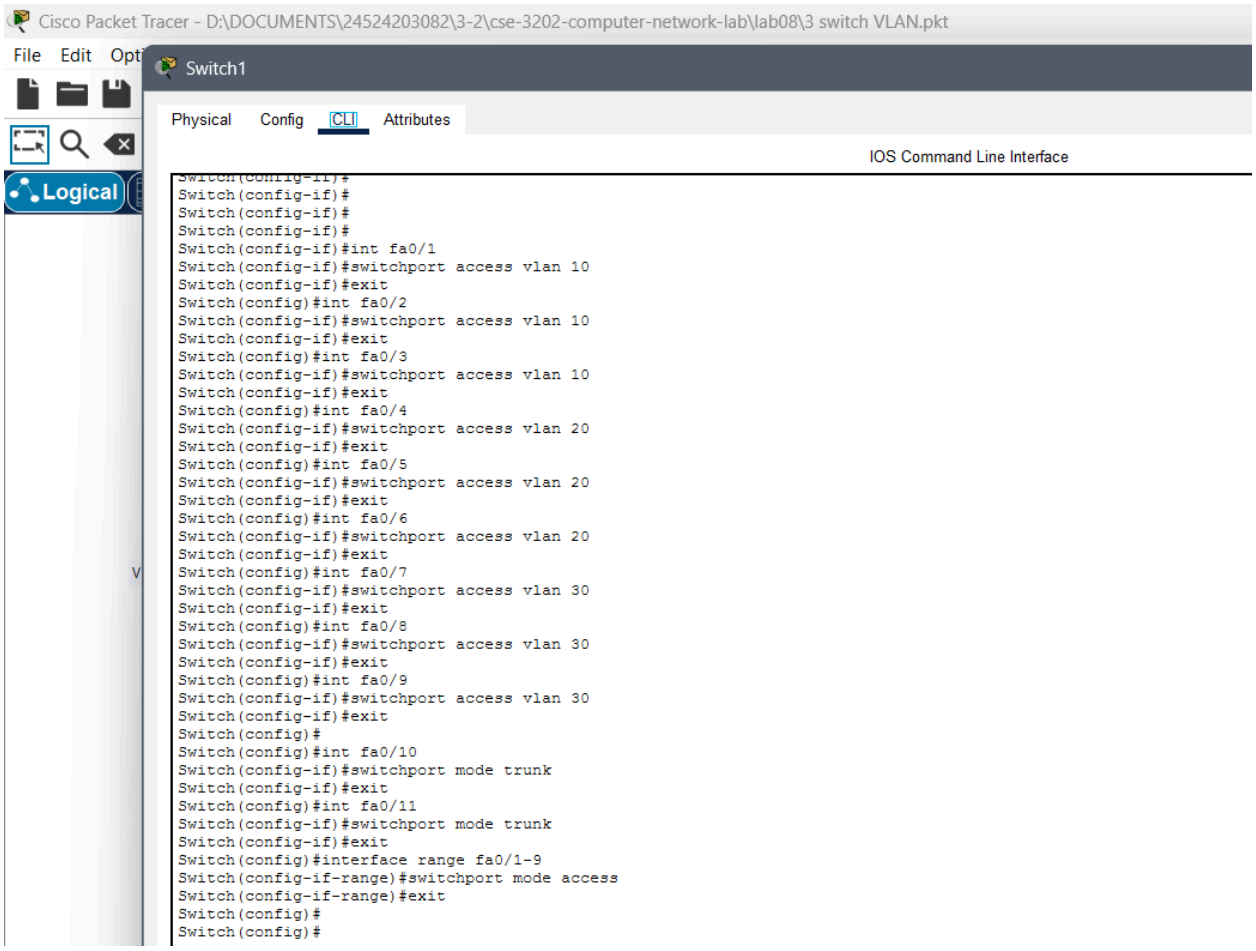


Figure 3: Port Configuration on Switch

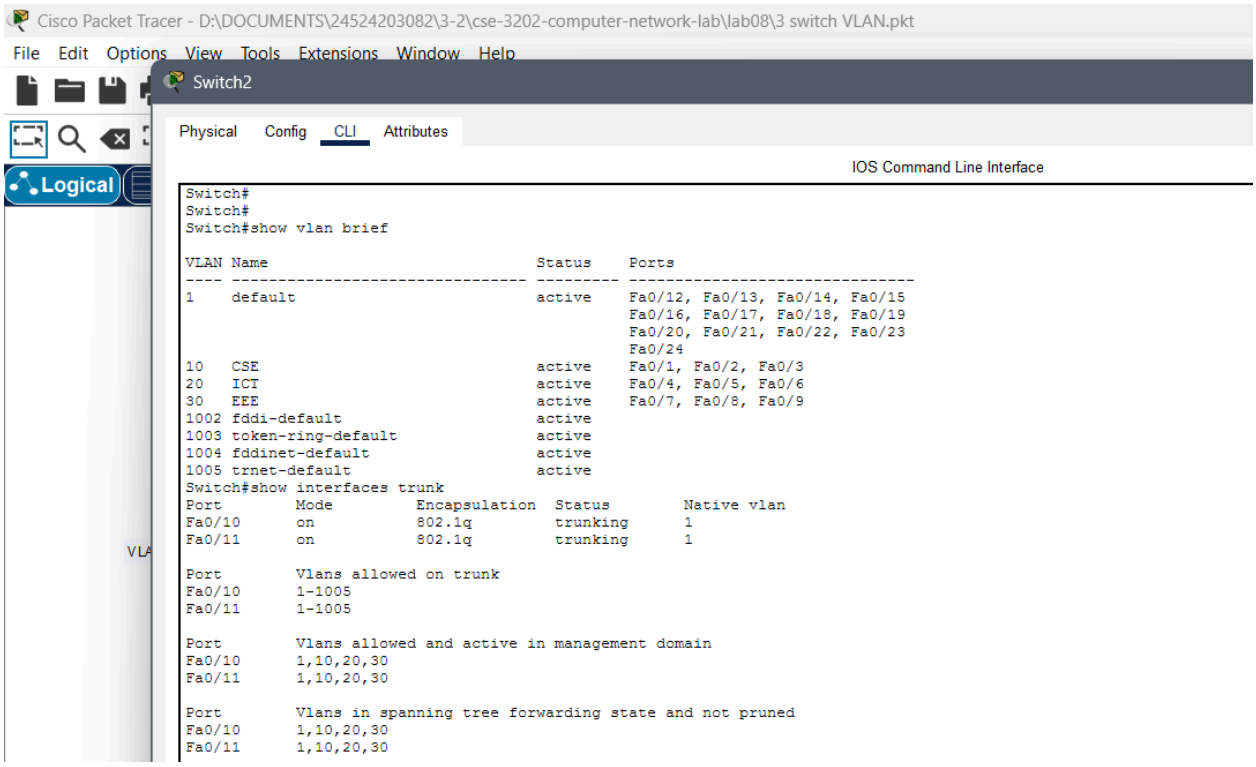


Figure 4: VLAN and TRUNK Verifications

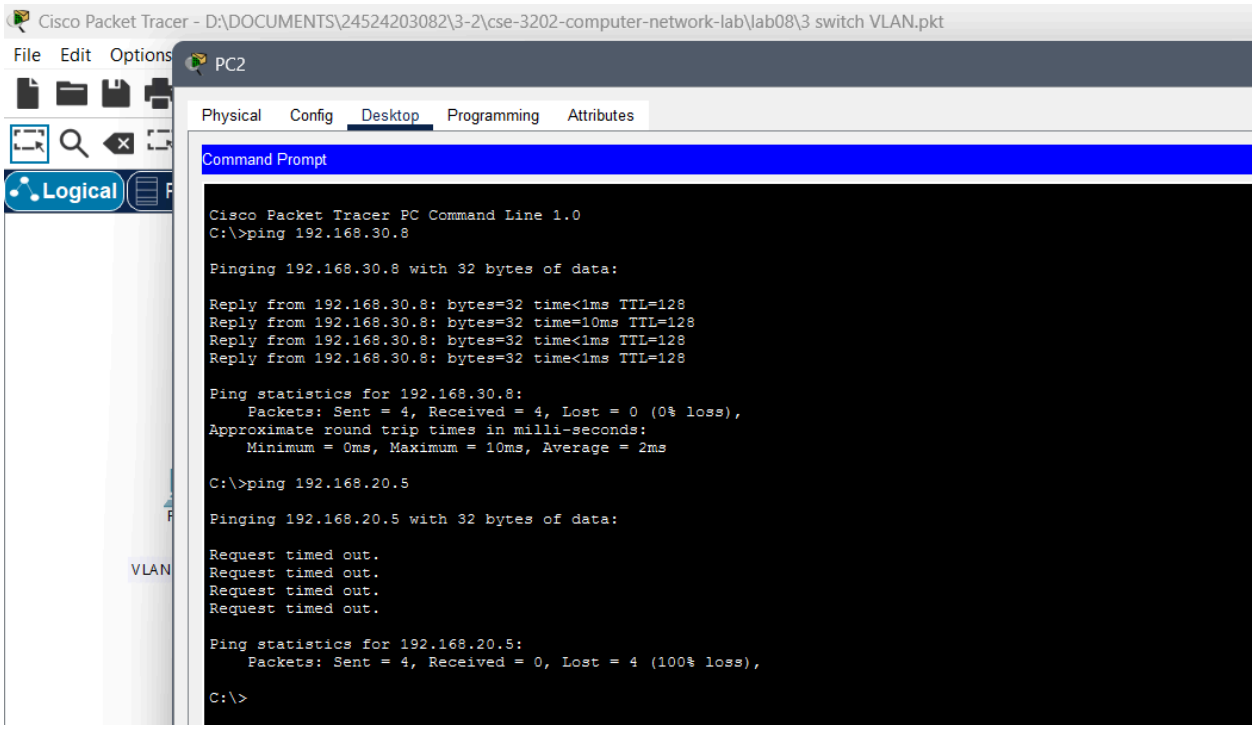


Figure 5: Ping Between PCs in the Same and Different VLANs